

1 a) ~~Symmetric~~, Positive Definite, Orthogonal

~~Diagonal~~

$$b) H A^* H^T = B \quad H (A + I)^{-1} H^T = C$$

c) Positive Semidefinite, Symmetric, Singular
Symmetric, Podef

$$2 \quad a) -1, 0, 1 \quad \begin{bmatrix} -1 & 0 & 2 & 4 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow V = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$b) \cancel{S} A \cancel{S}^{-1} = S \Lambda^{1001} S^{-1} = A^{1001}$$

$$c) 0, 2\lambda S$$

$$3 \quad a) \frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots \quad A q_j = \lambda_j q_j \rightarrow A^{-1} q_j = \frac{1}{\lambda_j} q_j$$

$$b) Q^T b$$

$$c) \frac{C_1}{\lambda_1}$$